

THE EFFECTS OF PLYOMETRIC TRAINING ON SPRINT PERFORMANCE: A META-ANALYSIS

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ABSTRACT

Sáez de Villarreal, E, Requena, B, and Cronin, JB. The effects of plyometric training on sprint performance. A meta-analysis. *J Strength Cond Res* 26(2): 575–584, 2012—The purpose of this meta-analysis was to attempt to gain a clear picture of the magnitude of sprint performance improvements expected after chronic plyometric training (PT) and to identify specific factors that influence the treatment effects. Studies employing a PT intervention and containing data necessary to calculate effect size (ES) were included in the analysis. A total of 26 studies with a total of 56 ES met the inclusion criterion. Analysis of ES demonstrated that the strategies that seem to maximize the probability of obtaining significantly ($p < 0.05$) greater improvement in sprint performance included training volume for <10 weeks; a minimum of 15 sessions; and high-intensity programs with >80 combined jumps per session. To optimize sprint enhancement, the combination of different types of plyometrics and the use of training programs that incorporate greater horizontal acceleration (i.e., sprint-specific plyometric exercises, jumps with horizontal displacement) would be recommended, rather than using only one form of jump training ($p < 0.05$). No extra benefits were found to be gained from doing plyometrics with added weight. The loading parameters identified in this analysis should be considered by the professional sprinters and specialized trainers with regard to the most appropriate dose-response trends PT to optimized sprint performance gains.

KEY WORDS velocity, effect size, lower limb, jumps

INTRODUCTION

Improving sprint performance is beneficial to many sports for a multitude of reasons from winning a race to providing an advantage during sprint duels that allow players to reach the ball before the opponent

(53). Because of the advantage of having greater speed, a great deal of research has focused on the development of sprint performance using a myriad of training methods, including speed training, sprint drills, sprinting against resistances, weight training, combined resistance and speed training, and plyometric training (PT) (17,18,31,42,46). The focus of this article is on the effects of PT on sprint performance. Plyometrics refers to exercises that are designed to enhance muscle, mainly through the use of jump training. Plyometric exercises constitute a natural part of most sport movements because they involve jumping, hopping, and skipping (i.e., such as high jumping, throwing, or kicking) (3–5). The identifying feature of plyometric exercise is a lengthening (eccentric contraction) of the muscle-tendon unit followed directly by a shortening or concentric contraction, otherwise termed a stretch-shortening cycle (SSC). The SSC is integral to plyometric exercise because it enhances the ability of the muscle-tendon unit to produce maximal force in the shortest amount of time.

The plyometric exercises used in a training program should match the individual needs of the athlete in relation to the characteristics of the sporting activity that they are involved with. That is, to optimize transference to sport plyometric exercises should reflect the type of activity implicit in that sport, that is, the principle of specificity. For example, jumping exercises that were nonspecific to running performance (i.e., vertical-type jump exercise) did not cause any effect on running speed (16,21). When exercises were specific (e.g., speed bounding) to running performance, the training program had a positive effect on running velocity (42).

Plyometric exercises can either be combined within a training program (combination training) or can be used as standalone exercises. Furthermore, plyometrics can be performed at various intensity levels, ranging from low-intensity hops to high unilateral-intensity drills such as bounding (alternating single-leg jumps for maximum horizontal distance). As far as the lower body is concerned, plyometrics include the performance of various types of body-weight jumping exercises, such as the drop jump (DJ) or countermovement jump (CMJ), alternate-leg bounding, hopping, and other SSC jumping exercises (7,8,10,12).

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TABLE 1. Summary of characteristics of all studies meeting the inclusion criteria.*

Authors	Gr	Year	Treatm	Age		W	H	Exp	K Ply	Fit	TG	ES	D		BH	NoJ	NoE	Tp ly	R	JL	Test
				n	G (y)								Freq	wk							
Adams et al. (2)	E	1987	Plyo	31	M 13			NR	N	G	-0.11	0.13	3	10	Low	56	30	1	DJ	N	Sprint 50 m
Chelly et al. (13)	E	2010	Plyo	12	M 19.1	70.3	176	Reg	N	G	-0.41	0.64	2	8	Mod	40	10	2	DJ	60	N Sprint 40 m
Chelly et al. (13)	C	2010		11	M 19	70.6	174	Reg	N	G	-0.11	0.18									Sprint 40 m
Chimera et al. (14)	E	2004	Plyo	9	F 20	59.2	164.5	Nat	N	G	-0.21	0.58	2	6	Mod		150	5	C	120	N Sprint 36.57 m
Chimera et al. (14)	C	2004		9	F 20	59.7	165.6	Nat		G	-0.19	0.48									Sprint 36.57 m
Delecluse et al. (17)	E	1995	Plyo	24	M 20	70.2	177		N	N	-0.21	0.37	2	9	Low		50	10	C		N Sprint 100 m
Delecluse et al. (17)	C	1995		15	M 20	70.2	177		N	N	-0.02	0.03									Sprint 100 m
Dodd et al. (20)	E	2007	Ply + WT	32	M 20.5			Nat	N	G	-0.02	0.22	3	12	Mod		36	3	C	180	N Sprint 18.28 m
Dodd et al. (20)	E	2007	Ply + WT	32	M 20.5			Nat	N	G	-0.01	0.06	3	12	Mod		36	3	C	180	N Sprint 36.57 m
Dodd et al. (20)	E	2007	Ply + WT	32	M 20.5			Nat	N	G	-0.02	0.07	3	12	Mod		36	3	C	180	N Sprint 54.86 m
Ford et al. (21)	E	1983	Plyo	12	M 17			NA	N	N	-0.17	0.41	3	10	Low	60	15	1	DJ	30	Y Sprint 36.57 m
Ford et al. (21)	E	1983	Ply + WT	15	M 17			NA	N	N	-0.11	0.22	3	10	Low	60	15	1	DJ	30	Y Sprint 36.57 m
Fry et al. (22)	E	1991	Ply + WT	14	F 19.6	64.3	171.9	Nat	N	E	0.15	0.39	2	12	Mod			7	C		N Sprint 9.1 m
Fry et al. (22)	E	1991	Ply + WT	14	F 19.6	64.3	171.9	Nat	N	E	-0.05	0.15	2	12	Mod			7	C		N Sprint 36.57 m
Herrero et al. (26)	E	2006	Plyo	11	M 20.8	79.7	179	NA	N	N	0.04	0.09	2	4	H		100		C	180	N Sprint 20 m
Herrero et al. (26)	E	2006	Ply + EMS	9	M 21.4	80.2	179	NA	N	N	-0.08	0.26	4	4	H		100		C	180	N Sprint 20 m
Herrero et al. (26)	C	2006		10	M 20.6	71.6	177				0.02	-0.10									Sprint 20 m
Herrero et al. (27)	E	2010	Ply + WT + EMS	11	M 21.4	80.2	179	NA	N	N	-0.01	0.12	4	4	Low				C		N Sprint 20 m
Herrero et al. (27)	E	2010	Ply + WT	8	M 20.9	79	172	NA	N	N	-0.04	0.41	4	4	Low				C		N Sprint 20 m
Herrero et al. (27)	C	2010		10	M 20.6	71.6	177	NA	N	N	0.02	-0.14									Sprint 20 m
Impellizzeri et al. (29)	E	2008	Plyo	22	M 25	74	178	Nat	Y	G	-0.07	0.77	3	4	H		125	4	C	30	N Sprint 10 m
Impellizzeri et al. (29)	E	2008	Plyo	22	M 25	74	178	Nat	Y	G	-0.09	1.01	3	4	H		125	4	C	30	N Sprint 20 m
Impellizzeri et al. (29)	E	2008	Plyo	22	M 25	74	178	Nat	Y	G	-0.08	0.79	3	4	H		125	4	C	30	N Sprint 10 m
Impellizzeri et al. (29)	E	2008	Plyo	22	M 25	74	178	Nat	Y	G	-0.08	0.47	3	4	H		125	4	C	30	N Sprint 20 m
Kotzamanidis (30)	E	2006	Plyo	15	M 11.1	49.6	156.8	NA	N	N	-0.14	0.38	2	10	Mod	20	80		C	180	N Sprint 30 m
Kotzamanidis (30)	C	2006		15	M 10.9	48.7	154.2	NA	N	N	0.03	-0.18									Sprint 30 m
Lyttle et al. (32)	E	1996	Plyo	11	M 23.9	79.1	182.3	Reg	N	G	0.07	0.16	2	8	Mod		40	1	SJ		N Sprint 40 m
Lyttle et al. (32)	E	1996	Ply + WT	11	M 23.8	72.5	178.4	Reg	N	G	-0.04	0.16	2	8	Mod	20	10	1	DJ		N Sprint 40 m
Lyttle et al. (32)	C	1996		11	M 20.6	74.2	177.3	Reg		G	-0.03	0.12									Sprint 40 m
Markovic et al. (33)	E	2007	Plyo	30	M 20.1	76.7	181	Nat	N	G	-0.07	0.20	3	10	H	60	70	1	DJ	180	N Sprint 20 m
Markovic et al. (33)	C	2007		33	M 20.1	76.7	181	Nat	N	G	0.04	-0.12									Sprint 20 m
Meylan et al. (36)	E	2009	Plyo	14	M 13.3	48.6	159	Reg	N	G	-0.04	0.52	2	8	H		150	4	C	90	N Sprint 10 m
Meylan et al. (36)	C	2009		11	M 13.1	47.4	163	Reg	N	G	-0.05	0.36									Sprint 10 m
Moore et al. (37)	E	2005	Plyo	7	B 20.2			Nat	Y	G	-0.18	0.37	3	12	Mod		175	6	C		N Sprint 25 m
Polhemus et al. (38)	E	1980	Ply + WT	27	M 26			NR	N	N	-0.33	0.76	3	6	Mod	25	35	2	S + D	60	Y Sprint 40 m
Rimmer et al. (42)	E	2000	Plyo	10	M 24	83	177	NR	N	G	-0.10	0.32	2	8	H		117	4	C	180	N Sprint 40 m
Rimmer et al. (42)	C	2000		9	M 24	83	177	NR	N	G	0.02	-0.09									Sprint 40 m
Robinson et al. (43)	E	2004	Plyo	15	F 20.6	68.5	167.2	NA	N	N	-0.41	0.92	3	8	Low		80	10	C		N Sprint 40 m
Rønnestad et al. (44)	E	2008	Ply + WT	8	M 23	73.5	180	Nat	N	G	-0.06	0.93	2	7	Low		50	3	C	60	N Sprint 40 m

(Continued on next page)

Sáez-Saez de Villarreal et al. (46)	E	2008 Plyo	42 M	22.3	75.6	174.7	NA	N	N	−0.03	0.21	1	7	H	60	60	1	DJ	60	N	Sprint 20 m
Sáez-Saez de Villarreal et al. (46)	E	2008 Plyo	42 M	23.1	80.1	176.6	NA	N	N	−0.03	0.23	2	7	H	60	60	1	DJ	60	N	Sprint 20 m
Sáez-Saez de Villarreal et al. (46)	E	2008 Plyo	42 M	21.8	72.68	175.5	NA	N	N	−0.07	0.85	4	7	H	60	60	1	DJ	60	N	Sprint 20 m
Sáez-Saez de Villarreal et al. (46)	C	2008	42 M	23.6	78.56	180.3	NA	N	N	−0.01	0.03										Sprint 20 m
Sáez-Saez de Villarreal et al. (47)	E	2010 Plyo	11 F	40.1	66.6	165.1	NA	N	B	−0.01	0.12	3	8	Low		150	4	C	180	N	Sprint 10 m
Sáez-Saez de Villarreal et al. (47)	E	2010 Plyo	20 F	53.2	71.2	163.3	NA	N	B	−0.01	0.22	3	8	Low		150	4	C	180	N	Sprint 10 m
Sáez-Saez de Villarreal et al. (47)	E	2010 Plyo	24 F	64.8	72.6	162.7	NA	N	B	0.02	−0.10	3	8	Low		150	4	C	180	N	Sprint 10 m
Salonikidis et al. (50)	E	2008 Plyo	16 M	21.1	71.7	174	Reg	N	N	−0.03	0.12	3	9	Mod		44	6	C	120	N	Sprint 12 m
Salonikidis and Zafeiridis (50)	C	2008	16 M	21.1	71.7	174	Reg	N	N	0.02	−0.15										Sprint 12 m
Siegler et al. (52)	E	2003 Ply+WT	34 F	16.5	61.46	167.4	Nat	N	N	−0.10	0.62	3	10	Low		50	3	C		N	Sprint 20 m
Siegler et al. (52)	C	2003	34 F	16.3	58	166.7	Nat	N	N	−0.04	0.19										Sprint 20 m
Thomas et al. (55)	E	2009 Plyo	12 M	17.3	68.7	177.9	Nat	N	G	−0.03	0.22	2	6	H	40	120	1	DJ		N	Sprint 20 m
Thomas et al. (55)	E	2009 Plyo	12 M	17.3	68.7	177.9	Nat	N	G	−0.02	0.09	2	6	H		120	1	C		N	Sprint 20 m
Wagner and Kocak (58)	E	1997 Plyo	60 M	17.5			Reg	N	N	−0.16	0.68	3	6	H		80	4	C		N	Sprint 50 m
Wagner and Kocak (58)	C	1997	60 M	17.5			NA	N	B	−0.01	0.02										Sprint 50 m
Wilson et al. (61)	E	1993 Plyo	13 M	22.1	71.6	174	NR	Y	N	−0.01	0.12	2	10	Mod	20	60	2	DJ	180	N	Sprint 30 m
Wilson et al. (61)	C	1993	13 M	24.1	76.1	173	NR		N	0.04	−0.06										Sprint 30 m

*Gr = group; E = experimental; C = control; G = gender; M = male; F = female; B = both; W = weight (kilograms); H = height (centimeters); Kply = knows plyometric; N = No; Y = yes; Treatm = treatment; Plyo = plyometric; Ply + WT = weight training; Ply + EMS = electrostimulation; JL = jump loaded; Y = Yes; N = No; NR = not reported; Freq = frequency days per weeks; D wk = duration weeks; Int = intensity; H = high; Mod = moderate; BH = box height (centimeters); NoJ = number of jumps; Tply = type of plyometrics; C = combined; S + D = SJ + DJ; R = rest (seconds); TG = time gains (seconds); NoE = no of exercises; SJ = squat jump; DJ = drop jump; Test = performance test; exp = experience; Nat = national; Reg = regional; NA = no athlete; NR = not reported; Fit = fitness; E = elite; G = good; N = normal, B = bad.

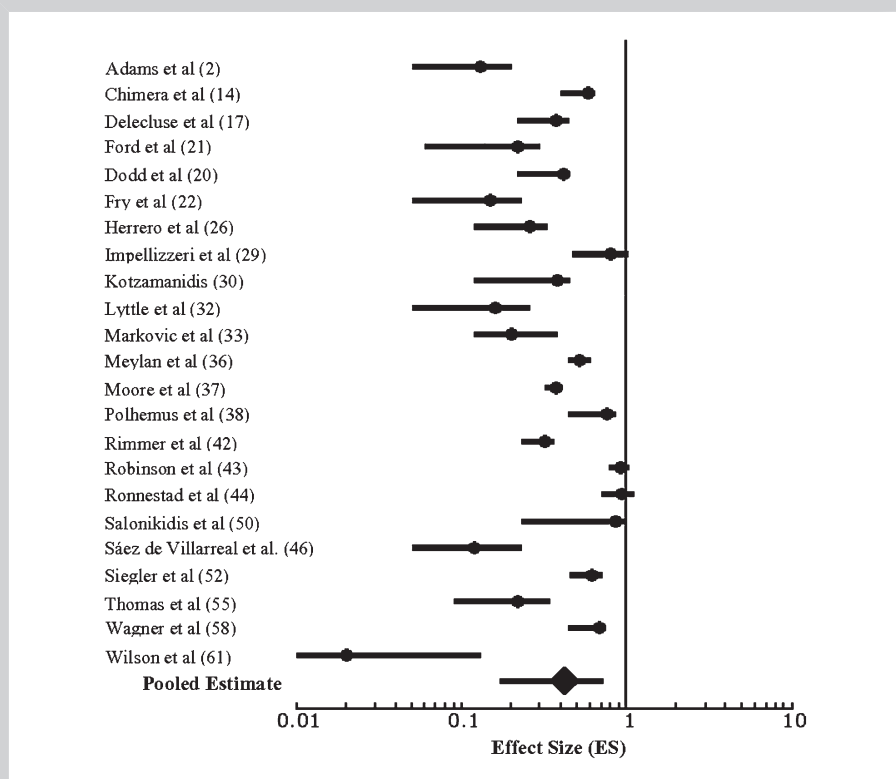


Figure 1. Effect size (ES) of all studies meeting the inclusion criteria. Horizontal bars represent 95% confidence intervals.

Although athletes and coaches involved in sprint training continue to use plyometric exercises (16,32), there are few studies describing the transfer of the training effects from PT in the horizontal plane to sprint performance in the acceleration phases of a sprint. The findings from the small number of studies in the literature regarding the effects of plyometrics on sprinting are inconsistent. Researchers have reported some improvements (18,42,46) and no improvements (22,32,61) in sprint times resulting from plyometric interventions. The plyometric exercises employed in these studies were not specific to sprinting, however, and the lack of specificity of the exercises to sprinting may have been responsible for the absence or small improvements in sprint times. On the other hand, improvements in 10- and 100-m sprint times have been found after a training intervention that incorporated some sprint-specific plyometric exercises (17).

The effects of PT may differ depending on the various subjects' characteristics, such as strength training level (14,20,32), gender (22,29), age (19,30,33,36,55), sport activity, or familiarity with PT (17,37). Research studies that combine these variables in different ways sometimes lead to conflicting results (14,30,32). Other factors that seem to determine the effectiveness of PT are program duration, type of plyometrics (i.e., vertical- or horizontal-type jump exercise), training intensity, or volume. Researchers have used numerous

combinations of these variables (20,26,46,61); therefore, the optimal combination of these factors for maximum achievement remains unclear.

The principal issue of determining the optimal loading parameters of a plyometric program remains inconclusive. Using a meta-analytical approach may lend some clarity to this area. Meta-analysis is a quantitative approach in which individual study findings addressing a common problem are statistically integrated and analyzed (25,41) overcoming the problems associated with small sample size and low statistical power. Because meta-analyses can effectively increase the overall sample size, it can also provide a more precise estimate of the effect of plyometrics on sprint performance. In addition, meta-analyses can account for the factors partly responsible for the variability in treatment effects observed among different

training studies (22,29,30,46,61). Thus, the purpose of this review was to examine the influence of various factors on the effectiveness of PT using a meta-analytical approach.

METHODS

Experimental Approach to the Problem

In the present study, the meta-analysis was performed in different steps, grounded in previous recommendations (48).

Subjects

A search was performed using key words in the English, French, and Spanish languages (e.g., jump training, sprint training, sprint performance, sprint times, SSC, plyometric, plyometrics, training of power, PT, pliometrique, entrainement pliometrique, salto pliométrico, velocidad). These key words were applied in the databases ADONIS, ERIC, SPORTSDiscus, EBSCOhost, MedLine, and PubMed. Moreover, manual searches of relevant journals and reference lists obtained from articles were conducted. The present meta-analysis includes studies published in journals that have presented original research data on healthy human subjects. No age, gender, or language restrictions were imposed during the search stage.

Research studies implementing PT programs for lower-limb muscles were the primary focus so research investigating

TABLE 2. Analysis for independent variables of subject characteristics.*

Subject characteristics								
Independent variables	Average (s) $\pm$ SD	<i>F</i>	Level	ES	SD	<i>n</i>	<i>r</i>	<i>p</i>
Age (y)						41	0.221	0.353
Body mass (kg)						32	−0.104	0.636
Height (cm)						32	−0.107	0.626
Group size						41	0.170	0.353
Previous experience		<i>F</i> (1,41) = 2.847	<i>p</i> = 0.102					
Familiarized	−0.08 $\pm$ 0.11			0.31	0.33	6		
Not familiarized	−0.08 $\pm$ 0.05			0.57	0.35	35		
Fitness		<i>F</i> (2,41) = 2.275	<i>p</i> = 0.121					
Normal	−0.11 $\pm$ 0.12			0.39	0.31	17		
Good	−0.08 $\pm$ 0.07			0.39	0.33	19		
Elite	0.05 $\pm$ 0.14			−0.12	0.38	5		
Gender		<i>F</i> (2,41) = 0.004	<i>p</i> = 0.996					
Male	−0.07 $\pm$ 0.08			0.36	0.32	32		
Female	−0.12 $\pm$ 0.20			0.37	0.50	8		
Both	−0.18 $\pm$ 0.00			0.37		1		
Sport level		<i>F</i> (2,36) = 0.555	<i>p</i> = 0.581					
National	−0.06 $\pm$ 0.08			0.41	0.39	16		
Regional	−0.04 $\pm$ 0.09			0.20	0.35	6		
No athletes	−0.11 $\pm$ 0.12			0.37	0.32	14		
Sport activity		<i>F</i> (11,38) = 6.310	<i>p</i> = 0.002†					
Volleyball	0.05 $\pm$ 0.00			−0.12	0.03	2		
Basket	−0.16 $\pm$ 0.00			0.68		1		
Body building	−0.01 $\pm$ 0.00			0.02		1		
Football	−0.10 $\pm$ 0.05			0.69†	0.22	12		
Swimming	0.01 $\pm$ 0.07			−0.01	0.00	2		
Phys. ed. stud.	−0.11 $\pm$ 0.12			0.35	0.30	11		
American football	−0.13 $\pm$ 0.00			0.13		1		
Baseball	−0.01 $\pm$ 0.00			0.11	0.08	3		
Rugby	−0.10 $\pm$ 0.00			0.32		1		
Tennis	−0.03 $\pm$ 0.00			0.12		1		
Track and field	−0.33 $\pm$ 0.00			0.76		1		
No athletes	−0.14 $\pm$ 0.00			0.38	0.01	2		

*ES = effect size; *n* = sample; level = alpha level; *r* = Pearson correlation coefficient; *p* = alpha level.

†*p* < 0.05.

the training of the upper-limb musculature was rejected. A total of 33 studies were initially identified.

The next step was to select studies with respect to their internal validity. Selection was based on the recommendations by Campbell and Stanley (11) and included (a) randomized control studies; (b) studies using instruments with high reliability and validity; (c) studies with minimal experimental mortality; (d) studies where the plyometric program was described; and (e) studies where the sprint test was conducted preintervention and postintervention. Twenty-six studies were selected after having completed all quality conditions (2,13,14,17,20–22,26,27,29,30,32,33,36–38,42–44,46,47,50,52,55,58,61) (Table 1).

Each study was read and coded independently by 2 investigators using different moderator variables. Because of the high number of variables that may affect training efficacy,

independent variables were grouped into the following areas: (a) subject characteristics: variables included age (years), body mass (kilograms), height (centimeters), previous experience, group size, level of fitness, sports level, and type of sport activity; (b) program exercises: variables included combination with other types of exercise, intensity of session, type of plyometric exercises and resistance; (c) program elements: variables included frequency of weekly sessions, program duration, drop height, number of jumps per session, number of exercises per session, and rest intervals between series of exercises; and (d) outcome measurements the type of sprint test used to identify performance gains (e.g., 50, 40, 30 m). The mean agreement was calculated by intraclass correlation coefficient. The coding agreement between investigators was determined by dividing the variables coded the same by the total number of variables. A mean agreement of 0.90 is accepted as an appropriate level of reliability for such coding

TABLE 3. Analysis of variance results on the differences of ES between various elements of plyometric training independent variables of program elements.*

Program exercises						
Independent variables	Average (s) $\pm$ SD	F	Level	ES	SD	n
Combination with other types of exercise		$F(2,40) = 0.501$	$p = 0.611$			
Plyometric	-0.10 ± 0.10			0.40	0.33	28
Ply+ weight training	-0.05 ± 0.11			0.28	0.38	11
Ply+ electrostimulation	-0.08 ± 0.10			0.26		1
Intensity of session		$F(2,41) = 3.684$	$p = 0.058$			
High	-0.17 ± 0.11			0.51	0.32	15
Moderate	-0.06 ± 0.12			0.20	0.29	12
Low	-0.06 ± 0.14			0.27	0.33	14
Type of plyometric exercises		$F(3,41) = 4.512$	$p = 0.009^\dagger$			
Combined	-0.09 ± 0.11			0.41 [†]	0.36	28
Squat jump	0.07 ± 0.00			-0.16		1
Drop jump	-0.07 ± 0.05			0.27	0.24	11
SJ + DJ	-0.33 ± 0.00			0.76 [†]		1
Resistance		$F(1,41) = 0.270$	$p = 0.607$			
Added weight	-0.20 ± 0.11			0.46	0.27	3
Weightless	-0.07 ± 0.10			0.35	0.35	38

*SJ = squat jump; DJ = drop jump; Ply = plyometric; ES = effect size; n = sample; level = alpha level.
[†] $p < 0.05$.

procedures (37). The mean agreement between coding for this study was 0.90. Any coding differences between investigators were scrutinized and resolved apriori to the analysis.

The effect size (ES) is a standardized value that permits the determination of the magnitude of the differences between the groups or experimental conditions (54). Gain ESs were calculated using Hedges and Olkin's g (25), using the formula

$$g = (M_{\text{post}} - M_{\text{pre}}) / SD_{\text{pooled}}, \quad (1)$$

where M_{post} is the mean for the posttest, M_{pre} is the mean for the pretest, and SD_{pooled} is the pooled SD of the measurements:

$$SD_{\text{pooled}} = \frac{(M_{\text{post}} - M_{\text{pre}})}{\sqrt{((n_1 - 1)SD_1^2 + (n_2 - 1)SD_2^2) / (n_1 + n_2 - 2)}}. \quad (2)$$

It has been suggested (45,48,54) that ES should be corrected for the magnitude of sample size of each study. Therefore, correction was performed using the formula

$$1 - 3 / (4m - 9), \quad (3)$$

where $m = n - 1$, as proposed by Hedges and Olkin (25).

Statistical Analyses

To examine the effect of the categorical independent variables on the ES, an analysis of variance (ANOVA) was used (23,45,56). In the case of quantitative independent variables (e.g., age, height,

duration of the treatment in weeks, number of jump per session), a Pearson's (r) correlation test was used to examine the relationships between ESs and variable values (45). Statistical significance was set at $p \leq 0.05$ for all analyses. The scale used for interpretation was the one proposed by Rhea (39,40), which is specific to training research and the training status of the subjects to evaluate the relative magnitude of an ES. The magnitudes of the ESs were considered either trivial (<0.35), small ($0.35-0.80$), moderate ($0.80-1.50$), or large (>1.5).

TABLE 4. Pearson correlation coefficients (r) between various program elements and training gains.*

Training program variables	n	r	p
Frequency session per wk	41	0.362	0.042 [†]
Program duration (wk)	41	-0.505	0.003 [‡]
Drop height (cm)	13	0.041	0.904
Number of jumps per session	37	0.367	0.040 [†]
Number of exercise per session	36	0.151	0.434
Rest between sets (s)	27	-0.663	0.001 [‡]

*n = sample; r = Pearson correlation coefficient; p = alpha level.
[†] $p < 0.05$.
[‡] $p < 0.01$.

TABLE 5. Analysis for independent variables of outcome measurement.*

Independent variables	Outcome measurement					
	Average (s) $\pm$ SD	<i>F</i>	Level	ES	SD	<i>n</i>
Sprint test (m)		$F(5,41) = 0.141$	$p = 0.981$			
100	-0.21 ± 0.00			0.37		1
50	-0.10 ± 0.07			0.29	0.33	3
40	-0.12 ± 0.14			0.39	0.36	12
30	-0.07 ± 0.09			0.20	0.25	2
20	-0.06 ± 0.05			0.39	0.32	15
10	-0.007 ± 0.002			0.32	0.56	8

*ES = effect size; *n* = sample; level = alpha level.† $p < 0.05$.

RESULTS

The analysis showed that the average ES of the PT group (0.37; $n = 41$; 20.081 seconds) was significantly higher ($p < 0.05$) compared with the ES of controls (0.03; $n = 15$; -0.013 seconds).

With regards to the subjects' characteristics, the results indicate no significant correlation coefficient for age ($r = 0.221$), body mass ($r = -0.104$), height ($r = -0.107$), and group size ($r = 0.170$), with the magnitude of the ES (Table 2). The ANOVA showed no significant effects in any of the variables measured (i.e., previous experience, fitness level, gender and sport level). However, there is a significant effect ($p < 0.05$) in the variable sport activity.

The ANOVA showed no differences in ESs regarding the intensity of session, the different combinations of PT or among programs with and without added resistance. However, significant differences ($p < 0.05$) were found among the type of plyometric exercises (Table 3).

There was a relationship ($p < 0.05$) between the frequency session per week ($r = 0.362$), program duration (weeks) ($r = -0.505$) and rest between sets ($r = -0.663$) with PT ES, but no significant effects were found between drop height (centimeters) ($r = 0.041$), number of jumps per session ($r = 0.347$), and number of exercises per session ($r = -0.151$) with the PT ES (Table 4). No differences in ES ($p > 0.05$) were found among the different sprint tests (Table 5).

DISCUSSION

The current metanalysis support those of previous studies (2,14,17,21,30) that have concluded that PT seems to be an effective training method for the improvement of the sprint performance (ES = 0.37, i.e., plyometric group). Thus, the reported sprint time gains of > -0.081 seconds resulting from PT could be of practical relevance for trained athletes in

sports (i.e., short-distance sprints, high-intensity sprints, initial accelerations, specific explosive actions). The present meta-analysis offers robust quantitative evidence to this conclusion and provides some valuable information concerning the importance of controlling some determinant variables for the improvement of the performance.

The results of the meta-analysis indicate that when subjects can adequately follow plyometric exercises (e.g., technical

ability and adequate musculoskeletal strength), the training gains are independent of the fitness level (i.e., maximal aerobic power). Some authors explain that the gains that can be made via PT are dependent on training status (6). For example, Delecluse (17) studied beginner and experienced sprinters and found greater sprint increases for the beginner and smaller gains for the experienced sprinters. Furthermore, a major part of the improvements in untrained subjects during the initial weeks in power-type strength training is probably because of adaptations of the neural system (1,14,24,28,60).

Sprint performance improvement was found to be non-significantly greater when plyometrics was performed combined with other types of exercises (i.e., plyometric + weight training or plyometric + electrostimulation) than when performed alone (Table 3). There is a possibility that the subjects in the combination training group were exposed to a higher volume of training than those in the others groups, that is, the total workload was not equated between groups. It would be very interesting if future studies made an attempt to equate workloads between groups when comparing different training methods. Another difference is the model used to provide the training stimulus to the subjects. Training intensity, volume, and exercise selection followed the principle of progressive overload, starting with lower intensities, single-joint exercises, and less complex exercise techniques, and progressing to higher intensities, multijoint exercises, and more complex techniques. In any case, the optimal training strategy to enhance sprint performance appears to be a specific PT. That is, sprint performance gains will be optimized by the use of training programs that incorporate greater horizontal acceleration (i.e., skipping, jumps with horizontal displacement). It is generally accepted that the more specific a training exercise is to a competitive movement, the greater is the transfer of the training effect to

performance (17,49). Athletes, such as sprinters, who require power for moving in the horizontal plane engage in bounding plyometric exercises, whereas athletes, such as high jumpers and volleyball players, who require power to be exerted in the vertical direction train using vertical jumping exercises (15,59).

It can also be concluded that when plyometric exercise intensity is high during the session, there is a greater improvement in sprint performance (Table 3). Some authors (9,51,57,63) determined that plyometric or SSC loading is higher during DJs, followed by CMJs, and then during squat jumps (SJs), there is little SSC loading. This is mainly attributed to the different characteristics of movement and, thus, to the different use of SSC characteristics. For these reasons, the combination of various exercises may result in higher gains compared with the performance of each exercise alone. Furthermore, it was found that a combination of plyometric exercises (i.e., SJ + DJ, bounding + CMJ) resulted in better training effects ($ESs = 0.76$) compared with the use of a single type of exercise (DJ) ($ESs = 0.27$) (Table 3). The higher improvements in sprint performance may be because of a training specificity. It is possible that a training program incorporating more horizontal acceleration (e.g., bounding and form running) may improve sprint times. In fact, it has been reported that no significant increase in sprint acceleration and velocity has resulted from training programs involving essentially vertical plyometric exercises (2,22,61). In contrast, significant improvements in running velocity, as measured by a 40-yd sprint, have resulted from the use of form running in conjunction with weighted depth jumps (21,38). Besides, improvements in 10- and 100-m sprint times have been found after a training intervention that incorporated some sprint-specific plyometric exercises (17).

Some research studies have shown that the PT with additional weights (vests, bars on the back, etc.) resulted in greater gains in sprint performance (21,38). The results of the meta-analysis indicated no significant differences among the training conditions (Table 3). Intuitively, this would make sense because adding weights increases ground contact times, and therefore, the duration of the eccentric and concentric phases are longer. Given that the magnitude of reflex potentiation, storage, and use of elastic energy is related to movement velocity and the time between the end of the eccentric phase and beginning of the concentric phase (coupling time), once more, the use of weights would seem problematic in the plyometric exercises and the sprints (i.e., the contact times) (42). The contact times during the initial acceleration phase of a sprint are similar to the contact times of the exercises employed (34,35,62,63). Therefore, the greatest transfer of the plyometrics to sprinting likely occurred during the initial acceleration phase. This theory is supported by Young (62), who suggested that bounding may be considered a specific exercise for the development of acceleration because of the similar contact times of bounding and sprinting during the

initial acceleration phase. Further research is required to test the theory that the greatest transfer of sprint-specific plyometrics to sprinting occurs during the phase of the sprint when the contact times of the sprint during that phase are the same as the contact times of the plyometric exercises. Therefore, training effects using additional weights are not guaranteed.

Volume and frequency are very important parameters to be taken into account for an optimum PT program design. The results showed that training for <10 weeks (i.e., between 6 and 8 weeks) with 3–4 sessions per week is more beneficial than similar programs of a longer duration. Similarly, treatment with more than 18 sessions increases sprint performance, whereas performance of >80 jumps per session seemed to result in the most beneficial volume (Table 4).

The finding of this study is that a sprint-specific plyometrics training program can improve sprint performance over distances down to 40 m in length. The effects of a sprint-specific plyometrics program appear to be the greatest over the initial meters (10–40 m). The results suggest that sports participants who are accustomed to performing sprints over distances up to 40 m could potentially improve sprint speed, particularly in the initial acceleration phase, by adding sprint-specific plyometric exercises to their training. Explosive speed is required in many sports and physical activities; coaches and participants should therefore consider a plyometrics training program that incorporates sprint-specific exercises as part of the overall training plan.

In conclusion, the present meta-analysis demonstrates that PT significantly improves sprint performance. The estimated improvements in velocity as a result of PT could be considered as practically relevant—for example, an improvement in sprint time of >–0.081 seconds (i.e., $ES = 0.37$) could be of high importance for trained athletes in sports relying on sprint performance. A training volume of <10 weeks (with >18 sessions) using high intensities (with >80 jumps per session) is the strategy that will maximize one's probability of obtaining significant improvements in performance. Another important conclusion is that sprint performance gains will be optimized by the use of training programs that incorporate greater horizontal acceleration (i.e., sprint-specific plyometric exercises, jumps with horizontal displacement). However, there are no extra benefits gained from doing plyometrics with added weight.

PRACTICAL APPLICATIONS

Plyometrics can be recommended as an effective form of physical conditioning for augmenting the sprint performance; yet, the effects of PT could vary because of a large number of variables, such as program duration, training volume, or intensity. The velocity and conditioning coach may consider taking into account the dose-response trends identified in this analysis to prescribe the appropriate level of training. Therefore, in addition to the well-known training methods such as resistance training, explosive and sprint training,

strength and conditioning professionals may well incorporate PT into an overall conditioning program of athletes striving to achieve a high level of explosive leg power and dynamic athletic performance.

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